	NATIONAL INSTITUTE OF TECHNOLOGY ANDHRA PRADESH **DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING** **II B.Tech II Semester Mid Examinations, February 2025**

CS255 – Database Management Systems

Date: 26-02-2025 **Time: 10:30 AM TO 12:30 PM** **Max. Marks: 30**

1 a) Consider the schema of student registrations and grades with example rows indicated. 2

StudentRegistrations(RegID, StudentID, AYear, Asem, ProgramYS, CourseCode)

2296, 1397, 2023, 1, 11, CE101

Course(CourseCode, CourseName, Credits,Type)

CE101, Engineering Mechanics,4, Theory

Grades(RegID, Marks, Grade)

2296 73 B

GradePoints(Grade, Points)

B 8.0

Identify source of redundancy in this schema and suggest how to mitigate it.

b) Suggest suitable extensions for the schema in 1 a) to incorporate courses across multiple regulations, courses taught by faculty from different departments and same course code offered to multiple sections to enable physical location of classes in multiple rooms. 4

2 Considering the employee table with the following structure

emp(empno, empname, hiredate, salary, mgr, job) answer the queries below:

a) Find all the employee and their manager names with employee manager name printed as "No-MGR" when there is no manager for the corresponding employee 1

b) Obtain the employees working under managers younger to them 1

c) List top-3 employees in each department in descending order of their salary 1

d) Report the salary and difference of salary of the particular employee and top-most earning employee's salary with in the same department 1.5

e) Report the pairs of departments where the salary of every employee within the department is greater than the average salary of employees of other department. 1.5

4 Consider the following relation schema:

ProductSupply (PID, SID, WID, S_Date, Quantity, S_Name, Warehouse_Location)

The following Functional Dependencies (FDs) hold:

PID, SID → S_Date, Quantity

SID → S_Name

WID → Warehouse_Location

PID, WID → Quantity

PID, SID, WID → S_Date, Quantity

a) Identify the candidate key for this schema. 2

b) Determine if the relation is in BCNF or not. If it is not, suggest a decomposition that satisfies BCNF conditions 4

3 a) A multi-specialty hospital wants to design a database to manage its operations, including patients, doctors, nurses, rooms, treatments, and medical records. The hospital has various departments (like neurology, ophthalmology, ortho, nephrology, etc..) and each department has specialized doctors working under it. Consider the following constraints on the design of schema for this workflow and evolve an E-R diagram that reflects the constraints. 6

A. Each patient must be assigned to one primary doctor but can also consult multiple doctors for specialized treatments.
B. Each doctor belongs to one department and may handle multiple patients at a time.
C. Doctors can be of different types: General Physicians, Surgeons, Cardiologists, Neurologists, etc.
D. A nurse is assigned to multiple patients per shift, but each patient may have one or more nurses attending to them.
E. Each nurse is assigned to a specific department.
F. Each patient must be assigned a room (either shared ward or private room).
G. A shared ward can accommodate multiple patients, whereas a private room can only have one patient at a time.
H. Rooms are identified by a room number and categorized by type (ICU, General, VIP, Isolation, etc.).
I. Every patient has a medical record, which must be associated with exactly one patient.
J. Each medical record includes prescriptions, lab tests, and treatment history.
K. Patients undergo multiple treatments (e.g., Surgery, Physical Therapy, Chemotherapy, Dialysis). Each treatment has a cost, which must be included in the billing system.
L. A patient's total bill should be calculated based on the room charges, treatment charges, and consultation fees.
M. Some treatments involve surgery, which requires one lead surgeon and multiple assisting doctors.
N. A surgery must be scheduled in an operating room, and cannot happen without one assigned surgeon.
O. Emergency patients are admitted without prior registration and are assigned an emergency room immediately. After stabilization, they may be transferred to a ward/private room or discharged.
P. A patient must be discharged by their primary doctor before leaving the hospital. Some patients require follow-up visits, which must be scheduled before discharge. Discharge summary must be prepared which includes Diagnosis details, Final treatment provided, Medications prescribed for home care, Follow-up visit schedule (if required), Doctor's final remarks

Identify the strong entities and minimum of two weak entities with their identifying entities in this workflow and various relationships.

5 a) Write the queries for obtaining the CGPAs of students for the schema given in 1 a) 3

b) Suggest three uses of views, give a suitable example for each use 3